

Aaryan Ahlawat

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Computer Engineering graduate (GPA 3.9) with experience in FPGA/Verilog design, SDN/network engineering, embedded systems, and AI/ML pipelines. Proven under NASA-level documentation standards. Eager to contribute to ERCOT's grid & market systems. U.S. Citizen — eligible for security clearance.

Education

- University of California, Santa Cruz — B.S. Computer Engineering | GPA: 3.9** **Jun 2026**
- Dean's Honor List: Fall 2024, Winter 2025, Spring 2025 (Awarded to top students earning 3.8+ GPA)
 - Relevant Coursework: Signals & Systems, Digital Logic Design, OS, ML, Database Systems, Networks
- Mission College, Santa Clara — A.A. Natural Science & Mathematics | Cum Laude** **May 2024**
- Focus: Sustainability-focused systems, network security, mathematics, and computing foundations
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Projects & Research

- Power Consumption Monitoring System (Python, PowerBox API, MongoDB, InfluxDB)** **Dec 2025**
- Measured real-time power draw of Tor Snowflake via PowerBox API; stored time-series data in InfluxDB and metadata in MongoDB for cross-hardware comparative analysis
 - Established power heuristics and schema design applicable to grid metering and analytics workflows
- FPGA-Based Arcade Game Engine (Verilog, Vivado) — UCSC** **May 2025**
- Implemented structural Verilog game engine with pixel-accurate VGA (640x480 @ 60Hz); validated via Vivado testbenches and oscilloscope/DMM signal verification
 - Debugged positive-edge-triggered D flip-flop timing on breadboard discrete logic circuits using oscilloscope
- SDN Firewall & Router (Mininet, Python, OpenFlow/POX) — UCSC** **Mar 2025**
- Built SDN firewall using POX/OpenFlow 1.3 with stateful TCP/UDP/ICMP filtering across a 5-switch university topology serving 15+ hosts across Faculty, Student, IT, and Data Center LANs
 - Mitigated ping-flood DoS attack; analyzed TCP under 0/20/100% packet loss via Wireshark – parallels ERCOT's SCADA/network reliability requirements
- GUSynth: Real-Time Timbre Transfer System (PyTorch, DDSP) — UCSC Capstone** **Jun 2026**
- Built real-time guitar-to-piano timbre transfer pipeline: YIN f0 extraction, RMS loudness, and 20-coefficient MFCC features fed into a 4-layer MLP (512 hidden units) mapping to additive synth params
 - Implemented 64-harmonic additive synthesizer with phase-continuous output at ~11ms total latency; trained on paired audio data with PyTorch over two quarters of iterative evaluation
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Work Experience

- NASA L'SPACE Mission Concept Academy (MCA) — NASA, Phoenix, AZ** **Sep 2024 – Dec 2024**
- Deputy PM of Resources & Engineer for \$300M lunar mission; co-authored 30% of 120-page NASA PDR covering FMEA, systems architecture, and lifecycle validation
 - Applied NASA SDLC-equivalent standards; briefed senior NASA engineers — directly parallels ERCOT's grid/market application development lifecycle
- L'SPACE NASA NPWEE — NASA, Phoenix, AZ** **May 2024 – Jul 2024**
- Lead Engineer in 12-week NASA program; contributed to peer-reviewed AI research on ML-based solar activity prediction for Mars exploration
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Skills

Programming: Python, C/C++, Java, SQL (PL/SQL), Verilog, Shell Scripting, JavaScript, x86 Assembly
Systems & Tools: Linux, Windows, Vivado, Git, Mininet, Packet Tracer, Oscilloscope, DMMs, FPGA, Azure
AI/ML & Testing: PyTorch/TensorFlow, LangChain, HuggingFace, Pandas, NLP, pytest, CI/CD, FMEA
Concepts: Data Structures, SDN/OpenFlow, REST APIs, SCADA (coursework), RISC-V, State Machines, BGP, VLANs